

Headache Classification

Overview

Headache is one of the most common neurological symptoms experienced by people of all ages. A headache is defined as pain or discomfort in the head, scalp, or neck. While many headaches are harmless and temporary, some may indicate a serious underlying medical condition requiring urgent evaluation.

Headaches are broadly classified into **primary headaches**, where the headache itself is the main disorder, and **secondary headaches**, where the headache is caused by another medical condition such as infection, stroke, trauma, or a brain tumor.

Proper classification of headaches helps healthcare professionals determine the underlying cause and select appropriate treatment. A detailed medical history, neurological examination, and, when necessary, imaging studies are important in the diagnostic process.

Types of Headaches

Headaches are divided into two major categories.

Primary Headaches

Primary headaches are not caused by another disease. They result from overactivity of pain-sensitive structures in the brain and surrounding tissues.

Common primary headaches include:

- Migraine
- Tension-type headache
- Cluster headache

Primary headaches are generally not life-threatening but may significantly affect a person's quality of life.

Secondary Headaches

Secondary headaches occur due to another underlying medical condition.

Common causes include:

- Stroke
- Brain tumor
- Meningitis
- Head injury
- Sinus infection
- High blood pressure
- Brain hemorrhage
- Medication overuse
- Hydrocephalus

Secondary headaches may require urgent medical attention depending on the cause.

Tension-Type Headache

Tension-type headache is the most common type of primary headache.

Characteristics

- Mild to moderate pain
- Pressure or tightness around the head
- Pain on both sides of the head
- Pain in the forehead, temples, or back of the head
- Neck and shoulder muscle tightness

Common Triggers

- Emotional stress
- Poor posture
- Lack of sleep
- Eye strain
- Anxiety
- Fatigue

Treatment

- Rest
 - Hydration
 - Stress management
 - Over-the-counter pain relievers
 - Regular exercise
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Migraine Headache

Migraine is a neurological disorder causing recurrent attacks of moderate to severe headache.

Characteristics

- One-sided headache
- Throbbing or pulsating pain
- Moderate to severe intensity
- Pain worsens with activity

Associated Symptoms

- Nausea
- Vomiting
- Sensitivity to light
- Sensitivity to sound
- Blurred vision
- Aura in some patients

Common Triggers

- Hormonal changes
- Stress
- Bright lights

- Sleep deprivation
 - Certain foods
 - Dehydration
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Cluster Headache

Cluster headache is a rare but extremely painful type of headache.

Characteristics

- Severe pain around one eye
- Sudden onset
- Lasts between 15 minutes and 3 hours
- Occurs in clusters over several weeks

Associated Symptoms

- Red eye
- Tearing
- Nasal congestion
- Runny nose
- Sweating on one side of the face
- Restlessness

Cluster headaches occur more commonly in men than women.

Secondary Headache Causes

Several medical conditions can cause secondary headaches.

Stroke

Headache associated with stroke may occur suddenly and may be accompanied by:

- Weakness
- Slurred speech

- Facial drooping
 - Vision changes
 - Difficulty walking
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Brain Tumor

Headache due to a brain tumor may:

- Become progressively worse
 - Be worse in the morning
 - Be associated with vomiting
 - Cause seizures
 - Cause weakness or vision problems
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Meningitis

Headache caused by meningitis is usually accompanied by:

- High fever
- Neck stiffness
- Confusion
- Sensitivity to light
- Vomiting

This is a medical emergency requiring immediate treatment.

Sinusitis

Sinus headaches may present with:

- Pain around the forehead
- Facial pressure
- Nasal congestion

- Fever
 - Thick nasal discharge
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Medication-Overuse Headache

Frequent use of pain-relieving medications can itself lead to chronic headaches.

Patients using pain medication on many days each month should seek medical advice.

Red Flag Symptoms

Certain headache symptoms suggest a potentially serious underlying condition.

Immediate medical evaluation is recommended if headache is associated with:

- Sudden severe headache ("worst headache of life")
- Sudden loss of consciousness
- Persistent vomiting
- High fever
- Neck stiffness
- Seizure
- Weakness of an arm or leg
- Facial drooping
- Slurred speech
- Confusion
- Vision loss
- Double vision
- Difficulty walking
- Recent head injury

These symptoms may indicate conditions such as stroke, brain hemorrhage, meningitis, or brain tumor.

Diagnosis

Evaluation of headache begins with a detailed medical history and physical examination.

Healthcare providers may assess:

- Headache location
- Duration
- Frequency
- Severity
- Triggering factors
- Associated symptoms
- Previous medical history
- Medication use
- Family history

Additional investigations may include:

- Neurological examination
- CT scan
- MRI
- Blood tests
- Lumbar puncture (when infection is suspected)
- Eye examination

Treatment

Treatment depends on the underlying type of headache.

Primary Headaches

Management may include:

- Pain-relieving medications

- Preventive medications
 - Stress reduction
 - Sleep improvement
 - Hydration
 - Lifestyle modification
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Secondary Headaches

Treatment focuses on the underlying cause.

Examples include:

- Antibiotics for meningitis
 - Surgery for brain tumors
 - Emergency treatment for stroke
 - Blood pressure control
 - Treatment of sinus infection
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Prevention

Many headaches can be prevented through healthy lifestyle habits.

Recommendations include:

- Maintain regular sleep schedules
- Stay hydrated
- Eat balanced meals
- Exercise regularly
- Reduce stress
- Avoid excessive caffeine
- Limit alcohol consumption
- Avoid smoking

- Maintain good posture
- Limit prolonged screen time

Patients with frequent headaches should maintain a headache diary to identify triggers and monitor treatment response.

Patient Advice

Most headaches improve with rest, hydration, and appropriate medication. However, headaches that suddenly change in pattern, become unusually severe, or are associated with neurological symptoms should never be ignored.

Patients experiencing frequent headaches should seek evaluation from a healthcare provider rather than relying on repeated self-medication. Early diagnosis can identify treatable conditions and improve long-term outcomes.

Summary

Headaches are broadly classified into primary and secondary types. Primary headaches include tension-type headaches, migraines, and cluster headaches, while secondary headaches result from underlying medical conditions such as stroke, brain tumors, infections, or trauma. Proper classification, early diagnosis, and appropriate treatment are essential for effective management. Recognizing red flag symptoms helps identify patients who require urgent medical attention.

Why this document is useful for your RAG system

This document intentionally includes common patient expressions that users may type into the chatbot, such as:

- "I have a severe headache"
- "My headache won't go away"
- "Pain around my eyes"
- "Pressure in my forehead"
- "One-sided headache"

- "Headache with vomiting"
- "Sudden worst headache of my life"
- "Headache with fever"
- "Headache with neck stiffness"
- "Headache after head injury"
- "Pain behind my eyes"
- "Daily headaches"

These phrases improve semantic retrieval, allowing your embedding model to match real-world patient questions with the appropriate headache classification information in your Neurology knowledge base.